

Introduction

Within the group project, I served as the **Communications and Fundraising Officer**. Initially, I saw this as a more outward-facing role, mainly focused on the device's presentation and possible funding sources. However, as the project progressed, I began to understand that communication in a medtech project cannot be separated from the technical side. In the early stages, we worked to understand different bone assessment modalities, including DEXA, CT, MRI, ultrasound, implant-based sensing, and vibration-based monitoring. This helped the group decide that vibration-based monitoring was the most suitable direction for our concept.

The overall team dynamic was very positive. Before the January break, we had already decided on our direction, and from that point onwards we worked steadily towards the final submission. This reflection examines how my role evolved, what I contributed, the challenges I faced, and what I learned through the process.

Specific Contributions

One of my main contributions was helping shape the outward-facing identity of the project. In collaboration with the Chief Designer and Market Researcher, I developed the logo, brand identity, and launch page for **OsteoTrack** so that the product had a more professional and recognisable online presence. Alongside this, I also researched intellectual property considerations, fundraising options, and early-stage startup planning.

As the project progressed, I began to realise that communication in a regulated medtech environment must be grounded in clinical reality. While developing our marketing strategy, I had to think beyond the product itself and understand the NHS workflow, particularly the difference between the patient as the user and the NHS as the economic buyer. This significantly changed the way I approached my role. Instead of only thinking about how to make the product appealing, I started thinking more seriously about how to present a technical idea in a way that was realistic, relevant, and investable.

What Went Well

The organisation and level of commitment within the group were among the strongest aspects of the project. Despite the pressure of other modules, deadlines, and responsibilities, we managed to maintain good momentum through around fifteen weekly meetings. At the end of each meeting, we set clear goals for the following week, which helped keep the project moving and ensured everyone stayed aligned.

This structure also helped when certain parts of the project depended on input from different team members. Rather than wasting time waiting, we continued progressing the sections that could already be developed. I am proud of how self-regulated the

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group was and how everyone delivered when it mattered. That consistency was a major reason why the final document felt coherent and well connected, rather than just a combination of separate pieces of work.

Challenges Faced

One challenge I faced was that my role depended on understanding technical details that were not originally my main responsibility. To contribute meaningfully to the fundraising pathway and wider commercial strategy, I first had to understand the science behind the device and the intended clinical workflow. This pushed me further into technical interpretation than I had expected, but it also strengthened my contribution because it meant that communication and commercialization were being developed alongside the engineering work rather than being added later.

Another challenge was aligning my own research-driven ideas with the broader direction of the group. Early on, I had concerns about how strong the evidence for vibration-based assessment was in real clinical settings, since much of the literature appeared to rely on modelled bones rather than actual patients. After researching competitor technologies, I suggested whether a resonance-based approach like **Ostell's implant technology** could be adapted for larger bones, possibly alongside fixation plates. However, after discussion, the group agreed that this would move too far away from our goal of creating a non-invasive device and would also make the regulatory pathway more complicated. Even though my idea was not pursued, I do not see that research as wasted. It helped us understand the competitor landscape better and made me realise that a strong technical idea still has to align with the wider clinical and regulatory aims of the project.

It was also difficult to balance ambition with realism. Conveying a vision for OsteoTrack was relatively easy, but presenting it in a way that felt believable for an early-stage concept was much harder. Because of this, I tried to keep the startup strategy grounded by focusing on realistic next steps such as prototype refinement, intellectual property protection, early validation, and pilot engagement, rather than presenting the product as more advanced than it really was.

What I Learned

From a team perspective, I learned how important it is for a group to work as a collective rather than as individuals trying to stand out separately. Since we were students from different backgrounds, it made sense for roles to develop naturally around each person's strengths. In my view, that was one of the main reasons the project worked as well as it did.

I also learned that marketing and fundraising are not secondary to innovation, but part of what makes innovation workable in the first place. A strong engineering idea needs

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more than enthusiasm or good branding to be presented effectively. It requires a real understanding of the technical basis, the clinical relevance, and the practical path to market. That was probably the most important lesson I learned from the entire process.

What I Would Do Differently

If I were to repeat the task, I would start the strategic communication side earlier and keep a more regular reflective log so I could monitor how our commercial thinking changed alongside the technical development. I would also create a clearer structure linking technical milestones to communication outputs from the outset. Rather than only assessing whether the strategy looked strong, I would test earlier whether it clearly addressed the needs of procurement teams, clinicians, and other key adopters.

Conclusion

Overall, I believe my contribution helped turn a technical concept into a more credible and outward-facing venture. My role developed from basic communication into broader work involving branding, market positioning, fundraising, and commercialization. I take great pride in the identity I helped create for **OsteoTrack**, and this project gave me a much deeper appreciation of how closely technical understanding and strategic thinking need to work together in medtech innovation.